

① Geography — Cave Features & Karst Topography

Q1. Choose the correctly matched cave feature and its formation process:

- (i) Stalactites — hang downward from cave ceiling formed by dripping water
- (ii) Stalagmites — rise upward from cave floor formed by dripping water
- (iii) Pillars — formed when stalactites and stalagmites meet

A Only (i) and (ii) are correctly matched

B Only (i) and (iii) are correctly matched

C Only (ii) and (iii) are correctly matched

D All of the above are correctly matched ✓

✓ **Answer: D — All three are correctly matched**

MEMORY TRICK

STALACTite → C for Ceiling (holds tight to top) | **STALAGMite** → G for Ground (grows upward)

Pillar / Column = Stalactite + Stalagmite grow toward each other and JOIN

Feature	Location	Formation
Stalactite	Hangs from CEILING	Calcite deposits as water drips from roof
Stalagmite	Rises from FLOOR	Calcite deposits as water hits floor
Pillar / Column	Connects ceiling & floor	Stalactite + Stalagmite meet

KARST TOPOGRAPHY — KEY FEATURES

Formed in **Limestone / Dolomitic rock** regions | India: Southern region

Features: Lapies → Limestone Pavement → Sinkhole/Doline → Uvala → **Caves (Stalactite + Stalagmite)**

Depositional features inside cave | Erosional features on surface



② History — Mughal Tombs & Hasht Bihisht Layout 🕌

Q2. Which of the following Mughal tombs **DEVIATES** from the typical square Hasht Bihisht (eight-part) layout, which consists of a central hall surrounded by eight rooms?

A Humayun's Tomb, Delhi

B Akbar's Tomb, Sikandra ✓

C Itmad-ud-Daulah's Tomb, Agra

D Safdarjung's Tomb, Delhi

✓ **Answer: B — Akbar's Tomb (Sikandra) deviates from standard Hasht Bihisht layout; it shows Persian, Jain & Hindu architectural influences reflecting his Sulh-e-Kul policy**

Tomb	Built By	Location	Key Points
Humayun's Tomb	Haji Begum / Bega Begum (wife)	Delhi	First Mughal architecture; UNESCO WHS; Charbagh style; Also called Aram Bagh = First Mughal garden; Hasht Bihisht layout

Akbar's Tomb	Jahangir (son)	Sikandra, Agra	DEVIATES from layout ; Persian + Jain + Hindu influence; Reflects Din-i-Ilahi / Sulh-e-Kul (religious tolerance)
Itmad-ud-Daulah	Nur Jahan / Mehr-un-Nissa	Agra	Called 'Baby Taj'; built in memory of her father; first use of pietra dura (inlay work)
Safdarjung's Tomb	Nawab of Awadh	Delhi	Last Mughal-style garden tomb; also follows Hasht Bihisht

EXTRA FACT — PANCHMAHAL

Panch Mahal, Fatehpur Sikri (Agra) → Built by Akbar → Architectural style: **Buddhist Vihara style** (5-storey structure like a Buddhist monastery)



③ Economy / Welfare — Social Security Schemes

Q3. Which government scheme aims to provide social security to workers in the unorganised sector?

A Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA)

B Atal Pension Yojana (APY) ✓

C Pradhan Mantri Jan Dhan Yojana (PMJDY)

D Pradhan Mantri Awas Yojana (PMAY)

✓ **Answer: B — Atal Pension Yojana (APY) is specifically a social security scheme for unorganised sector workers; MGNREGA is an employment guarantee scheme, not social security**

⚠ EXAM TRAP!

MGNREGA = Employment Guarantee (NOT social security!) | APY = Social Security for unorganised sector ✓

<i>Scheme</i>	<i>Type</i>	<i>Year</i>	<i>Ministry</i>	<i>Key Details</i>
MGNREGA	Employment Guarantee	Act: 2005 / Implemented: 2006 / MG NREGA: 2 Oct 2009	Rural Development	Now renamed BB Gram Ji Yojana (Dec 2025); 125 days from 1 July 2026 (was 100 days)
Atal Pension Yojana	Social Security ☒	2015	Finance	Regulatory body: PFRDA ; Age: 18–40 yrs; Pension after 60 yrs; ₹1,000–₹5,000/month; On death → spouse gets benefit
PM Jan Dhan Yojana	Financial Inclusion	2014	Finance	Zero-balance bank accounts; every household



④ Science — Plant Nutrients: Macronutrients vs Micronutrients

Q4. Which of the following pairs of nutrient types and examples are **CORRECTLY** matched?

- (i) Macronutrients — Carbon, Hydrogen, Oxygen, Nitrogen, Phosphorus, Potassium, Calcium, Magnesium, Sulfur
- (ii) Micronutrients — Iron (Fe), Manganese (Mn), Zinc (Zn), Copper (Cu), Boron (B)
- (iii) Macronutrients — Iron (Fe)

A Only (i) is correctly matched

B (i) and (ii) are correctly matched ✓

C Only (iii) is correctly matched

D All three are correctly matched

✓ **Answer: B** — Iron is a Micronutrient (not Macro). C, H, O, N, P, K, Ca, Mg, S are the 9 Macronutrients. Everything else (Fe, Mn, Zn, Cu etc.) = Micronutrients

PLANT MACRONUTRIENTS (9 total)

CHO-NPK-CaMgS

C-Carbon | H-Hydrogen | O-Oxygen

N-Nitrogen | P-Phosphorus | K-Potassium

Ca-Calcium | Mg-Magnesium | S-Sulfur

PLANT MICRONUTRIENTS

Fe-Iron ☒ | Mn-Manganese | Zn-Zinc

Cu-Copper | B-Boron | Mo-Molybdenum

Cl-Chlorine | Ni-Nickel

Needed in very small (trace) amounts

MEMORY TRICK

In PLANTS: CHO + NPK + CaMgS = 9 Macros | All others = Micros

In HUMANS: Macros = Proteins, Carbs, Fats | Micros = Vitamins & Minerals



⑤ Geography — Earth's Climate Zones & Seasons

Q5. Consider the following statements about the Earth's major climatic zones:

Statement 1: Tropical zones experience minimal seasonal temperature variation due to their proximity to the equator.

Statement 2: Temperate zones are characterised by four distinct seasons — Summer, Winter, Spring, and Autumn.

Which of the statements given above is/are correct?

A Statement 1 only

B Statement 2 only

C Both Statement 1 and Statement 2 ☒

D Neither Statement 1 nor Statement 2

☒ **Answer: C** — Both statements are correct. Tropical zone gets direct overhead sun rays → minimal seasonal variation. Temperate zones have 4 seasons due to Earth's axial tilt and revolution.

Zone	Latitude Range	Also Called	Key Feature
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Tropical	23.5°N (Tropic of Cancer) to 23.5°S (Tropic of Capricorn)	Torrid Zone	Direct overhead rays → Minimal seasonal temp variation
North Temperate	23.5°N to 66.5°N (Arctic Circle)	—	4 distinct seasons : Summer, Winter, Spring, Autumn/Fall
South Temperate	23.5°S to 66.5°S (Antarctic Circle)	—	
North Frigid	66.5°N to 90°N (North Pole)	Arctic Region	No direct overhead sun rays ever
South Frigid	66.5°S to 90°S (South Pole)	Antarctic Region	

FORMULA / RULE

KEY SUN DATES:

☀️ **June 21** → Sun overhead at Tropic of Cancer → Longest day N. Hemisphere → **Summer Solstice = International Yoga Day**

❄️ **December 22** → Sun overhead at Tropic of Capricorn → Shortest day N. Hemisphere → **Winter Solstice**

🌅 **March 21** → Sun overhead at Equator → Equal day & night worldwide → **Vernal / Spring Equinox**

🌆 **September 23** → Sun overhead at Equator → Equal day & night worldwide → **Autumnal Equinox**

WHY SEASONS OCCUR

1. Earth's **Revolution** around the Sun

2. Earth's **Axial Tilt** = 23.5° from its own axis = 66.5° from orbital plane

Continentality Effect: Places far from ocean → Extreme temperature variations

⚠️ EXAM TRAP!

Southern Hemisphere (Australia etc.) → Seasons are **OPPOSITE** to Northern Hemisphere!

When India has summer, Australia has winter and vice versa.



⑥ Art & Culture — Folk Dances & UNESCO

Q6. Regarding the Ghoomar dance of Rajasthan, consider the following statements:
 Statement 1: Bhil women were its earliest performers, and Rajputs adopted it later.
 Statement 2: Ghoomar was inscribed on the UNESCO Intangible Cultural Heritage List in 2015.
 Which of the above statements is/are correct?

A Statement 1 only

B Statement 2 only

C Both Statement 1 and Statement 2 ✓

D Neither Statement 1 nor Statement 2

✓ **Answer: C** — Both are correct. Ghoomar originally performed by Bhil tribal women; Rajputs later adopted it. UNESCO listed it in 2015 (not 2010 — that is Kalbelia!)

⚠ EXAM TRAP!

Ghoomar → UNESCO **2015** | Kalbelia → UNESCO **2010** → Don't confuse these years!

Dance	State / Community	UNESCO Year	Notes
Ghoomar	Rajasthan (Bhil → Rajputs)	2015	Circular dance; women in flowing ghagra
Kalbelia	Rajasthan (Kalbelia snake-charmer community)	2010	Serpentine dance movements
Jhumar	Haryana	—	—
Jhumur	Assam (tea garden communities)	—	—
Jhumair	Jharkhand (eastern belt)	—	—

UNESCO INTANGIBLE HERITAGE — INDIA (RECENT)

2010 → Kalbelia (Rajasthan) | **2015** → Ghoomar (Rajasthan)

2021 → Durga Puja, Kolkata | **2023** → Garba (Gujarat)

2025 → Diwali (Most Recent — Must Remember!)

⑦ History — Ancient Rulers & Administration

Q7. Which ancient ruler's administration is noted for its highly centralised structure, described in sources like Arthashastra and Mudrarakshasa, and also in accounts by the Greek ambassador Megasthenes?

A Harshavardhana

B Samudragupta

C Chandragupta Maurya ✓

D Pulakeshin II

✓ **Answer: C** — Chandragupta Maurya had the most centralised administration, described in Kautilya's Arthashastra. Megasthenes (sent by Seleucus Nicator) documented it in his book Indica.

Ruler	Dynasty / Capital	Period	Key Facts
Harshavardhana	Pushyabhuti; Capital: Thanesar → Kannauj	606– 647 AD	Court poet: Banabhatta (Harshacharita + Kadambari); His own books: Ratnavali, Priyadarshika, Nagananda; Defeated by Pulakeshin II (Battle of Narmada, 618 AD); No son → empire disintegrated; Kannauj Tripartite Struggle (Palas, Gurjara-Pratiharas, Rashtrakutas)
Samudragupta	Gupta	330– 380 AD	'Napoleon of India' (V.A. Smith); Court poet: Harisena (Prayag Prashasti); Played veena (shown on coins); 24 battles — never lost; Father: Chandragupta I; Son: Chandragupta II (Vikramaditya)
Chandragupta Maurya ✓	Maurya	~321– 297 BC	Moriya tribe (described in Puranas); Mudrarakshasa by Vishakhadatta; Kautilya/Chanakya → Arthashastra ; Megasthenes wrote Indica ; Defeated Seleucus Nicator; married Helena; performed Sallekhana at Shravanabelagola (Karnataka) with Bhadrabahu

Pulakeshin II	Chalukya; Capital: Badami / Vatapi	—	Defeated Harsha (Battle of Narmada); Defeated Mahendra Varman I (Pallava); Later defeated by Narasimha Varman I → who took title 'Vatapikonda'
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⑧ Polity — DPSP & Uniform Civil Code

Q8. Which part of the Indian Constitution includes the Directive Principles of State Policy (DPSP) regarding a Uniform Civil Code (Article 44)?

A Part III (Articles 12–35) — Fundamental Rights

B Part IV (Articles 36–51) — Directive Principles of State Policy ✓

C Part V (Articles 52–151) — The Union

D Part VI (Articles 152–237) — The States

✓ **Answer: B** — DPSP is in Part IV (Articles 36–51). Article 44 specifically deals with Uniform Civil Code (UCC)

DPSP TYPES (OUR CLASSIFICATION — NOT IN CONSTITUTION)

Liberal/Intellectual: Art 44 (UCC), Art 45 (Free Education), Art 48A (Environment)
Socialist: Art 39 (Equal pay), Art 41 (Right to work)
Gandhian: Art 40 (Village Panchayats), Art 46 (SC/ST welfare)

UCC — STATES IMPLEMENTING

1st State: **Uttarakhand**
 2nd State: **Gujarat**
 3rd State: **Assam** (May 2026)

FORMULA / RULE

PART V — THE UNION (5 Chapters):

Ch.1: Union Executive (President, VP, PM, CoM, Attorney General) | Art 52–78
 Ch.2: Parliament (Lok Sabha + Rajya Sabha + President) | Art 79–122

Ch.3: Legislative Powers of President — Ordinances | Art 123

Ch.4: Supreme Court | Art 124–147

Ch.5: CAG (Comptroller & Auditor General) | Art 148–151



⑨ GK — National Science Award & State Health Index 🏆

Q9. The National Science Award is a civilian science award. Which of the following statements is correct regarding this award?

A It is conferred by the Ministry of Culture

B It is conferred by the Ministry of Science and Technology ✓

C It is conferred by the Ministry of Education

D It is conferred by the Ministry of External Affairs

✓ **Answer: B — National Science Award → Ministry of Science and Technology. NOT Ministry of Culture (Culture gives cultural awards). If ministry is WRONG in options → entire statement is wrong!**

🧠 MEMORY TRICK

Exam Trick: If a question says "Science Award conferred by Ministry of Culture" → **WRONG** immediately!

Rule: Award type must match the Ministry. Science → Science Ministry | Culture → Culture Ministry

STATE HEALTH INDEX — KEY FACTS

Released by: **NITI Aayog + Ministry of Health & Family Welfare + World Bank** (Technical Support)

NOT CSIR, NOT Ministry of Earth Sciences, NOT Institute for Competitiveness

24 Health Indicators across 3 domains:

→ Health Outcomes | Governance & Information | Key Inputs & Processes

States ranked in 3 categories: Front Runner | Achiever | Aspirant

World Bank HQ: Washington D.C. | Formed: 1944 (Bretton Woods) | Original name: IBRD



⑩ Economy — Revenue Receipts & Assertion-Reason 💰

Q10. Assertion (A): The government's revenue receipts do not bring with them any burden of repayment of financial liability.

Reason (R): Revenue receipts come from tax-related sources and regular non-tax inflows, which are earnings and are not meant to be returned.

Choose the correct answer:

A Both A and R are true, and R is the correct explanation of A ✓

B Both A and R are true, but R is NOT the correct explanation of A

C A is true but R is false

D A is false but R is true

✓ **Answer: A — Both A and R are true. Revenue receipts (taxes + non-tax income) are permanent income — no repayment needed. R correctly explains why A is true.**

Head	Receipts	Expenditure
Revenue	Tax (Direct + Indirect) + Non-tax (interest, dividends). NO repayment burden	Salaries, pensions, subsidies, grants → No asset/liability created
Capital	Loans (liability created) + Disinvestment (asset lost) + Recovery of loans	Infrastructure/loans given (asset created) OR loan repayment (liability reduced)

📐 FORMULA / RULE

DIRECT TAX: Income Tax, Corporate Tax, Gift Tax, Wealth Tax (paid directly to Govt, can't avoid)

INDIRECT TAX: GST, Customs Duty, Excise Duty, Service Tax (collected via middlemen)

GST — KEY FACTS

101st Constitutional Amendment Act (Bill No. 122) | Implemented: 1 July 2017

Article 279A → GST Council (33 members)

Chairperson: **Union Finance Minister**

Voting: Centre = 1/3 | States together = 2/3 | Decisions by **3/4 majority**

Slabs: **0% | 5% | 12% | 18% | 28%**

FORMULA / RULE

DEFICIT TYPES:

Revenue Deficit = Revenue Expenditure – Revenue Receipts

Effective Revenue Deficit = Revenue Deficit – Grants to states for capital asset creation

Fiscal Deficit = Total Expenditure – Total Receipts (excluding borrowings)

Primary Deficit = Fiscal Deficit – Interest Payments

Budget Deficit = Total Expenditure – Total Receipts (everything included)



⑪ History — Temple Architecture Styles

Q11. Which of the following pairs of temple styles and their regions/characteristics are **CORRECTLY** matched?

- (i) Vesara style — North India
- (ii) Rekha Deula — Kalinga (Odisha) architecture
- (iii) Bhumija style — Malwa region (developed by Paramaras)
- (iv) Dravidian style — Tamil Nadu (Gopuram, Vimana)

A Only (i) and (ii) are correct

B (ii), (iii), and (iv) are correct ✓

C Only (iii) and (iv) are correct

D All four are correct

✓ **Answer: B** — Vesara is **NOT** North India; it is Deccan/Central India (developed by Chalukyas). Rekha Deula IS Kalinga (Odisha). Bhumija IS Malwa. Dravidian IS Tamil Nadu. ✓

Style	Region	Developed By	Key Features
Nagara (North Indian)	North India	Various N. Indian dynasties	Curvilinear Shikhara (tower); No Gopuram; Nandi inside temple
Dravidian	Tamil Nadu (South India)	Pallavas (founder: Simhavishnu; Capital: Kanchipuram)	Pyramidal Vimana + Gopuram (tall gateway) + Water Tank + Nandi OUTSIDE
Vesara	Deccan / Central India (NOT North India)	Chalukyas ; Hoysalas (stellate/star-shaped temples at Halebeedu/Dwara Samudra)	Mix of Nagara + Dravidian
Bhumija	Malwa (MP / Gujarat / Rajasthan)	Paramaras (Raja Bhoj); also Chalukyas	Miniature shikhara motifs
Kalinga	Odisha	Eastern Ganga dynasty (e.g., Narasimhadeva)	Rekha Deula (curvilinear main tower) + Pidha Deula (Jagamohana/Porch) + Pamsana Deula

PALLAVA TEMPLES — MUST KNOW

Pancha Rathas (Mahabalipuram) → Started by **Mahendra Varman I** / completed Narasimha Varman I

Shore Temple (Mahabalipuram) → **Narasimha Varman II (Rajasimha)**

Mahendra Varman I was DEFEATED by Pulakeshin II → his son Narasimha Varman I took revenge → titled '**Vatapikonda**'

CHOLA TEMPLES (UNESCO WHS — Great Living Chola Temples)

Brihadeeswara, **Tanjore** → **Raja Raja Chola**

Brihadeeswara, **Gangaikonda Cholapuram** → **Rajendra Chola**

Airavateshvara Temple, Darasuram → also UNESCO WHS



⑫ Polity — Parliament's Power to Legislate on State List

Q12. Under which of the following circumstances can Parliament legislate on a subject in the State List?

1. When the Rajya Sabha passes a resolution by special majority in national interest (Art 249)
2. When a National Emergency is proclaimed under Article 352 (Art 250)
3. When two or more states request Parliament to legislate on State List subject (Art 252)
4. To implement any international treaty or agreement (Art 253)
5. When President's Rule is imposed in a state (Art 356)

A Only 1, 2, and 3

B Only 2, 3, and 4

C Only 1, 2, 3, and 4

D All five — 1, 2, 3, 4, and 5 ✓

✓ **Answer: D** — All five situations allow Parliament to legislate on State List subjects. Laws under Art 249 are valid for 1 year; laws made under Art 356 (President's Rule) continue for 1 year after rule ends.

Article	Situation	Special Condition
Art 249	Rajya Sabha passes resolution → National Interest	Special majority (2/3 present + voting); valid for 1 year
Art 250	During National Emergency (Art 352)	Automatically applies during emergency
Art 252	Two or more states REQUEST Parliament to legislate	Only those states bound by it
Art 253	To implement International Treaties / Agreements / Conventions	Whole country bound
Art 356	President's Rule in a state	Laws remain valid for 1 year after President's Rule ends



13 Art & Culture — Kharchi Puja & Northeast Festivals 🎭👑

Q13. The Kharchi Puja of Tripura is associated with the worship of how many deities? It lasts for several days and involves ritual bathing to wash away sins, similar to taking a dip in the holy Ganges.

A 7 deities

B 10 deities

C 14 deities ✓

D 21 deities

✓ **Answer: C** — Kharchi Puja involves worship of 14 deities; lasts 7 days; celebrated in July in Tripura

⚠ EXAM TRAP!

Hojagiri (Tripura — Riang community) ≠ **Kojagiri** (Maharashtra). Very different festivals, don't mix them up!

Festival / Dance	State / Community	Key Details
Kharchi Puja	Tripura	14 deities; 7 days; celebrated in July; ritual bathing
Hojagiri	Tripura — Riang community	Dance festival ≠ Kojagiri (Maharashtra)
Garia Puja	Tripura / Northeast	Performed alongside Jhum (shifting) cultivation
Biju Festival	Chakma tribe (NE states)	New Year celebrations; Biju dance
Jhumur Dance	Assam (tea garden workers)	Harvest celebration

JHUM CULTIVATION

Shifting cultivation in Northeast India is called **Jhum** | Also known as slash-and-burn agriculture



⑭ History — Lahore Resolution 1940

Q14. The Lahore Resolution passed by the Muslim League in 1940 demanded which of the following?

- A** Complete integration of all princely states with British India
- B** Creation of a separate state for Muslims in the Northwest and Eastern parts of India ✓
- C** Equal representation for Muslims in all legislative bodies
- D** Continuation of a united India with enhanced safeguards for minority rights

✓ **Answer: B** — The Lahore Resolution (also called Pakistan Resolution) by Muslim League under Jinnah demanded a separate sovereign Muslim state — Pakistan — carved from NW (West Pakistan) and East (East Pakistan = present-day Bangladesh)

PARTITION TIMELINE

1940 — Lahore Resolution → Muslim League demands Pakistan

1947 — India–Pakistan Partition → West Pakistan + East Pakistan

1971 — India helps liberate Bangladesh (East Pakistan) from West Pakistan

Bangladesh War of Liberation: Indian military involvement under PM Indira Gandhi



⑮ Polity — Supreme Court Articles 124–151

Q15. Article 126 of the Indian Constitution mentions which of the following regarding the Supreme Court?

- A** Appointment of Ad hoc Judges to the Supreme Court
- B** Appointment of Acting Chief Justice of India ✓

c Supreme Court to be a court of record

d Seat of the Supreme Court

✓ **Answer: B — Article 126 = Appointment of Acting Chief Justice of India (when CJI is unable to perform duties, President appoints an acting CJI). Art 127 = Ad hoc judges. Art 129 = Court of record. Art 130 = Seat.**

Article	Subject
124	Establishment & Constitution of SC (commenced 28 Jan 1950)
125	Salaries & Allowances of SC Judges
126	Acting Chief Justice of India — President appoints when CJI unable to perform duties ✓
127	Ad hoc Judges — when shortage of quorum; CJI requests President; qualified HC judge appointed temporarily
128	Retired judges can be called back (with Presidential approval)
129	SC as Court of Record (judgments preserved; can punish for contempt of court)
130	Seat of SC = New Delhi; CJI can declare other seat with President's approval
131	Original Jurisdiction (EXCLUSIVE federal disputes: Centre vs State, State vs State)
132	Appellate Jurisdiction — Constitutional matters
133	Appellate Jurisdiction — Civil matters
134	Appellate Jurisdiction — Criminal matters
137	SC can review its OWN judgments/orders
141	Law declared by SC binding on ALL courts in India
142	SC can pass any decree/order for complete justice in any matter (basis of Judicial Overreach)
143	Advisory Jurisdiction — President can seek SC's opinion (not binding on President)
144	All civil & judicial authorities in India to act in aid of SC

SC JUDGE QUALIFICATIONS (Art 124)

Must be: HC Judge for **5 years**, OR Advocate in HC for **10 years**, OR **Distinguished Jurist** in President's view

Art 227 → HC has *Superintendence* over ALL courts & tribunals in state

Art 235 → HC has *Admin & Disciplinary Control* over subordinate courts

Art 144 → All courts act in AID of Supreme Court (national level)



⑩ Art & Culture — MP Festivals

Q16. During which festival do unmarried girls in Madhya Pradesh place lamps inside decorated urn pots (called "Gudliya") as part of the celebrations dedicated to Goddess Gauri (Parvati)?

A Bhagoria Haat Festival

B Gangaur Festival (Gudliya ritual) ✓

C Lokrang Festival

D Chetiyagiri Vihar Festival

✓ **Answer: B** — Gangaur Festival (March–April); Gudliya = placing lamp in decorated pot; dedicated to Goddess Gauri (Parvati); celebrated in Rajasthan and MP; unmarried girls pray for a good husband and happy married life

<i>Festival</i>	<i>State</i>	<i>Details</i>
Gangaur (Gudliya)	Rajasthan + MP	March–April; Goddess Gauri (Parvati); unmarried girls pray for good husband; lamp in decorated urn = Gudliya ✓
Bhagoria Haat	Madhya Pradesh	Tribal festival; associated with Bhil and Bhilala tribes; Holi time
Pola Festival	Madhya Pradesh / Maharashtra	Bullock worship festival
Lokrang	Madhya Pradesh (Bhopal)	Folk arts and cultural festival

Chetiagiri Vihar Festival	Sanchi (MP)	Buddhist festival; worship of relics of Mahamogallana & Sariputra at Sanchi Stupa
Khajuraho Dance Festival	Madhya Pradesh	Classical dance festival; Khajuraho temple backdrop



⑰ Science — Plant Biology: Tissues & Transport

Q17. Which of the following statements regarding plants is **INCORRECT**?

- A** Root hair cells increase the surface area for water absorption from the soil
- B** **Phloem tubes are specialised for transport of water and minerals throughout the plant ✓ (INCORRECT statement)**
- C** Chlorophyll, the green pigment in leaves, absorbs light energy for photosynthesis
- D** Stomata are tiny pores in leaves guarded by kidney-shaped guard cells that regulate gas exchange

✓ **Answer: B is the INCORRECT statement — Phloem transports FOOD (glucose), NOT water and minerals. XYLEM transports water and minerals. Option B is wrong, making it the answer to "which is incorrect".**

⚠ EXAM TRAP!

PHLOEM = Food transport | XYLEM = Water + Minerals transport. This is the most common exam confusion! Never swap these.

XYLEM (Complex Permanent Tissue)

Transports: **Water + Minerals** (upward)

Components: Tracheids, Vessels, Xylem

Fibres, Xylem Parenchyma

All **DEAD** cells **EXCEPT Xylem Parenchyma**

PHLOEM (Complex Permanent Tissue)

Transports: **Food / Glucose** (bidirectional)

Components: Sieve Tubes, Companion Cells,

Phloem Fibres, Phloem Parenchyma

All **LIVING** cells **EXCEPT Phloem Fibres**

PHOTOSYNTHESIS — LOCATION IN CHLOROPLAST

Plastids → Chloroplasts (food) | Chromoplasts (colour) | Leucoplasts (storage: amyloplast=carbs, elaioplast=fat, aleuroplast=protein)

Grana (Thylakoids) → Light reactions (Light-dependent) | **Stroma** → Calvin Cycle (Light-independent = Dark reactions)

Overall: Light energy → Chemical energy (food)



⑱ Literature — Famous Indian Authors & Their Works

Q18. The namesake is a famous novel by which Indian-American author, who is known for writing about the Bengali immigrant experience and whose other works include "The Namesake," "Interpreter of Maladies," "Unaccustomed Earth," and "The Lowland"?

A Arundhati Roy

B Kiran Desai

C Jhumpa Lahiri ✓

D Chitra Banerjee Divakaruni

✓ **Answer: C** — Jhumpa Lahiri is the Indian-American author (Bengali parents) who wrote *The Namesake*, *Interpreter of Maladies* (Pulitzer Prize), *Unaccustomed Earth*, *The Lowland*, and *Whereabouts*

Author	Famous Books	Awards / Notes
Jhumpa Lahiri	<i>The Namesake</i> , <i>Interpreter of Maladies</i> , <i>Unaccustomed Earth</i> , <i>The Lowland</i> , <i>Whereabouts</i>	Pulitzer Prize (<i>Interpreter of Maladies</i>); Indian-American; Bengali parents
Arundhati Roy	<i>The God of Small Things</i> , <i>The Ministry of Utmost Happiness</i>	First Indian to win Booker Prize (<i>God of Small Things</i> , 1997)

Kiran Desai	<i>The Inheritance of Loss</i>	Booker Prize 2006; daughter of Anita Desai
Anita Desai	<i>Fasting, Feasting, Clear Light of Day</i>	Kiran Desai's mother; German mother + Bengali father
Chitra Banerjee Divakaruni	<i>The Palace of Illusions, Mistress of Spices, The Forest of Enchantments</i>	—



⑩ Economy — National Income & Personal Income

Q19. Consider the following statements regarding income concepts in national accounting:

Statement 1: The total national income of a country is composed of two parts — Private Income and Public Income.

Statement 2: Out of private income, what finally reaches the household (after removing corporate taxes and undistributed profits) is called Personal Income.

Which of the statements given above is/are correct?

A Statement 1 only

B Statement 2 only

C Both Statement 1 and Statement 2 ✓

D Neither Statement 1 nor Statement 2

✓ Answer: C — Both are correct. Total National Income = Private Income + Public Income. Personal Income is derived from Private Income after removing corporate taxes and undistributed profits, and adding transfer payments.

FORMULA / RULE

NATIONAL INCOME → PERSONAL INCOME:

National Income – (Earned but not received: undistributed profits, retained earnings)
+ (Received but not earned: subsidies, scholarships, pensions, transfer payments)
= **Personal Income**

Personal Income – Direct Taxes = Personal Disposable Income

FOREIGN EXCHANGE RESERVES – COMPONENTS

1. **FCA** — Foreign Currency Assets (largest component)
2. **Gold**
3. **SDR** — Special Drawing Rights = "Paper Gold" = weighted average of **5 currencies: USD + EUR + GBP + JPY + CNY**
4. **RTP** — Reserve Tranche Position (at IMF)

1991 CRISIS — MUST KNOW

India had **only 2 weeks** of foreign exchange reserves → Devalued rupee twice:
July 1 → **9% devaluation** | July 3 → **11% devaluation**
Went to IMF for bailout loan → Pledged gold reserves



② Geography — Earthquake Waves & Volcanoes

Q20. Consider the following statements about earthquake waves:

Statement 1: P-waves (Primary waves) can travel through solids, liquids, and gases and are the first to be recorded on a seismograph.

Statement 2: S-waves (Secondary waves) can travel only through solid medium and are transverse in nature.

Statement 3: The shadow zone of the P-wave is larger than that of the S-wave.

Which of the above statements is/are correct?

A Statements 1 and 3 only

B Statements 1 and 2 only ✓

C Statements 2 and 3 only

D All three statements are correct

✓ **Answer: B** — Statement 3 is **WRONG**. The **S-wave shadow zone is LARGER** than P-wave shadow zone (S-wave covers ~40% of Earth's surface as shadow zone). Statements 1 & 2 are correct.

⚠ **EXAM TRAP!**

S-wave shadow zone is **LARGER** (~40% of Earth's surface), **NOT** P-wave. This is a common exam trap! P-wave shadow zone is smaller.

Wave Type	Category	Medium	Nature	Shadow Zone
P-Wave (Primary)	Body wave	Solid + Liquid + Gas	Longitudinal (compression & rarefaction); FASTEST	Smaller shadow zone
S-Wave (Secondary)	Body wave	Solid ONLY	Transverse (crest & trough); Slower than P	LARGER shadow zone (~40%)
Surface Waves	Surface wave	Earth's surface only	Love waves + Rayleigh waves	Most Destructive

KEY TERMS

Hypocentre / Focus = Origin of earthquake inside Earth

Epicentre = Point on surface directly above focus → Maximum destruction

Tsunami = Earthquake under sea bed → Japanese: "harbor waves"

VOLCANO TYPES

Caldera: Explosive; collapses on itself | **Composite/Stratovolcano**: Thick lava; steep

Shield: Fluid lava; gentle slopes (e.g., Hawaii: Mauna Kea, Mauna Loa)

Flood Basalt: Very fluid; large flat areas → **Deccan Plateau (India)** formed by flood basalt



21 Geography — Auroras

Q21. Which phenomenon is primarily responsible for the occurrence of auroras (polar lights) near the Earth's poles?

A Reflection of sunlight by polar ice caps

B Volcanic activity near the polar regions

C Interaction of solar wind particles with Earth's magnetic field and atmosphere ✓

D High concentration of ozone near the poles

✓ **Answer: C** — Solar wind (charged particles from the Sun) interacts with Earth's magnetosphere near the poles (where magnetic field lines converge) and excites atmospheric gases → Emit colourful light = Aurora

AURORA — KEY FACTS

Aurora Borealis = Northern Lights (North Pole / Arctic region)

Aurora Australis = Southern Lights (South Pole / Antarctic region)

Cause: Solar wind particles + Earth's magnetic field → Ionisation of atmosphere → Colourful light emission

Best viewed: Canada, Norway, Iceland, Alaska, New Zealand (south)



22 Sports — Badminton Court Dimensions

Q22. What is the standard width of a singles badminton court as per international (BWF) regulations?

A 6.1 metres (width of doubles court)

B 13.4 metres (length of full court)

C 5.18 metres ✓

D 12 metres

✓ **Answer: C** — Singles court width = 5.18 m. Full length = 13.4 m. Option A (6.1 m) is doubles court width — a common trap! Option B is the LENGTH not width.

Width of SINGLES = 5.18 m | Width of DOUBLES = 6.1 m | LENGTH (both) = 13.4 m →
Exam always tries to swap singles/doubles width!

Measurement	Value
Width — Singles Court	5.18 metres
Width — Doubles Court	6.1 metres
Length — Full Court (both)	13.4 metres
Net Height at Centre	5 feet (1.524 m)
Net Height at Edges/Posts	5 feet 1 inch (1.55 m)



23 Geography — Chambal Ravines

Q23. The Chambal Ravines are a notable geographical feature. In which of the following states are these ravines primarily found, and what type of erosion is primarily responsible for their formation?

A Gujarat and Maharashtra — Sheet erosion

B Rajasthan, Madhya Pradesh, and Uttar Pradesh — Ravine (Gully) erosion ✓

C Punjab and Haryana — Wind erosion

D Bihar and Jharkhand — Rill erosion

✓ **Answer: B** — Chambal ravines are found in Rajasthan, MP & UP. Formed by ravine/gully erosion. Chambal river originates in Mau Plateau (MP) and meets Yamuna near Etawah (UP).

Origin: **Mau Plateau (Madhya Pradesh)**

Flow: Rajasthan → MP → Uttar Pradesh

Meets: **Yamuna near Etawah (UP)**

Famous for: **Ravine/Gully erosion** | **Gharials** (*Gavialis gangeticus*) | National Chambal Sanctuary

Tributaries joining Yamuna: Chambal, Sindh, Betwa, Ken



24 History — Gupta Period Administration

Q24. Which of the following statements about the Mahasandhi Vighrahika during the Gupta period is **CORRECT**?

A He was the Chief Justice of the Kingdom and presided over land disputes

B He supervised irrigation works and agricultural revenue collection

C He was the Minister of War and Peace (Foreign Minister) who handled treaties and diplomatic relations ✓

D He managed temple affairs and religious ceremonies of the kingdom

✓ **Answer: C — Mahasandhi Vighrahika = Minister of War and Peace = Foreign Minister who handled treaties, diplomacy, and decisions of war and peace (Sandhi = peace/treaty, Vighraha = war)**

Gupta Officer	Role / Meaning
Mahasandhi Vighrahika	Minister of War & Peace = Foreign Minister ; arranged treaties (Sandhi=peace, Vighraha=war) ✓
Kumar Amatya	Most important officer; Provincial/Ministerial officer
Mahadanda Nayaka	Officer of Punishment / Chief Justice (Law enforcement)
Nagar Shresthi	Chief Banker of the city
Sarthavaha	Leader of merchant caravan / trade guild

Pratham Kulika	Chief Artisan / Head of craftsmen
Pratham Kayastha	Chief Scribe / Record keeper (like stenographer)
Vishayapati	Head of district (Vishaya = district in Gupta administration)

GUPTA TAX TERMS

Land Tax = **Udrang** | Emergency Tax = **Uparikara** | Gold Tax = **Hiranya**



Quick Revision — All 24 Questions at a Glance

Q#	Topic	Correct Answer
Q1	Cave Features (Karst Topography)	D — All correctly matched; Stalactite=Ceiling, Stalagmite=Floor, Pillar=Both meet
Q2	Mughal Tombs (Hasht Bihisht)	B — Akbar's Tomb (Sikandra) deviates; shows Persian+Jain+Hindu influence
Q3	Social Security Schemes	B — Atal Pension Yojana (APY); MGNREGA = Employment, NOT social security
Q4	Plant Macronutrients vs Micronutrients	B — CHO+NPK+CaMgS = Macros; Iron (Fe) = Micro, NOT Macro
Q5	Earth's Climate Zones	C — Both statements correct; Tropical=minimal variation; Temperate=4 seasons
Q6	Ghoomar Dance & UNESCO	C — Both correct; Bhil women first; UNESCO 2015 (NOT 2010=Kalbelia)
Q7	Ancient Rulers – Centralised Admin	C — Chandragupta Maurya (Arthashastra by Kautilya; Indica by Megasthenes)
Q8	DPSP & Uniform Civil Code (Art 44)	B — Part IV, Articles 36–51
Q9	National Science Award	B — Ministry of Science & Technology (NOT Culture)
Q10	Revenue Receipts A&R Question	A — Both A&R true; R correctly explains A; Revenue receipts = no repayment burden
Q11	Temple Architecture Styles	B — Vesara=Deccan NOT North India; Kalinga=Odisha; Bhumija=Malwa; Dravidian=TN

Q12	Parliament Legislating on State List	D — All 5 situations (Art 249, 250, 252, 253, 356)
Q13	Kharchi Puja (Tripura)	C — 14 deities; 7 days; July
Q14	Lahore Resolution 1940	B — Separate Muslim state in NW + Eastern India = Pakistan demand
Q15	Article 126 – Supreme Court	B — Acting Chief Justice of India (NOT Ad hoc=127, NOT Court of Record=129, NOT Seat=130)
Q16	MP Festivals (Gangaur/Gudliya)	B — Gangaur Festival; Gudliya = lamp in decorated urn by unmarried girls
Q17	Plant Biology (Incorrect Statement)	B — Phloem transports FOOD not water; Xylem transports water+minerals
Q18	Indian Authors & Their Books	C — Jhumpa Lahiri (The Namesake, Interpreter of Maladies etc.)
Q19	National Income Concepts	C — Both statements correct; National Income = Private + Public; Personal Income from Private Income
Q20	Earthquake Waves	B — St.1 & St.2 correct; St.3 WRONG (S-wave shadow zone is LARGER not P-wave)
Q21	Auroras (Polar Lights)	C — Solar wind + Earth's magnetic field interaction → Aurora Borealis (N) / Aurora Australis (S)
Q22	Badminton Court Dimensions	C — Singles width = 5.18 m (NOT 6.1m=doubles; NOT 13.4m=length)
Q23	Chambal Ravines	B — Rajasthan+MP+UP; Ravine/Gully erosion; meets Yamuna at Etawah
Q24	Gupta Period – Mahasandhi Vighrahika	C — Foreign Minister / Minister of War & Peace (NOT Chief Justice, NOT irrigation, NOT temple)



END OF LECTURE 12 — SSC GD ANSWER KEY NOTES

All 24 Questions ✓ | All Options Written ✓ | Full Explanations ✓

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